# Fractions, Decimals and Percentages

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**KEY WORDS** fraction *(hautanga)* · decimal *(tau ā-ira)* · percentage *(ōrau)* · per cent *(ōrau)* · tenths *(hautekau)*

## A Tenths and hundredths ●○○

⚠ **Watch out:** The first decimal place is tenths and the second is hundredths: $\frac{7}{10} = 0.7$ but $\frac{7}{100} = 0.07$.

**1.** Write each fraction as a decimal.

**a)** $\frac{3}{10}$ = _____ **b)** $\frac{7}{10}$ = _____ **c)** $\frac{9}{100}$ = _____ **d)** $\frac{23}{100}$ = _____

**e)** $\frac{1}{10}$ = _____ **f)** $\frac{61}{100}$ = _____ **g)** $\frac{1}{100}$ = _____ **h)** $\frac{99}{100}$ = _____

**2.** Write each decimal as a fraction in its simplest form.

**a)** 0.1 = _____ **b)** 0.7 = _____ **c)** 0.03 = _____ **d)** 0.47 = _____

**e)** 0.5 = _____ **f)** 0.2 = _____ **g)** 0.25 = _____ **h)** 0.75 = _____

## B Percentages ●●○

💡 **Remember:** Per cent means "out of 100". 37% is 37 out of 100, which is $\frac{37}{100}$ or 0.37.

**3.** Write each percentage as a decimal.

**a)** 50% = _____ **b)** 25% = _____ **c)** 70% = _____ **d)** 8% = _____

**e)** 99% = _____ **f)** 10% = _____ **g)** 3% = _____ **h)** 45% = _____

**4.** Write each decimal as a percentage.

**a)** 0.45 = _____ **b)** 0.7 = _____ **c)** 0.09 = _____ **d)** 0.3 = _____

**e)** 0.01 = _____ **f)** 0.85 = _____ **g)** 0.5 = _____ **h)** 0.25 = _____

## C Fractions and percentages ●●○

💡 **Remember:** Learn these: $\frac{1}{2} = 50\%$, $\frac{1}{4} = 25\%$, $\frac{3}{4} = 75\%$, $\frac{1}{10} = 10\%$ and $\frac{1}{5} = 20\%$.

**5.** Write each fraction as a percentage.

**a)** $\frac{1}{2}$ = _____ **b)** $\frac{1}{4}$ = _____ **c)** $\frac{1}{10}$ = _____ **d)** $\frac{3}{4}$ = _____

**e)** $\frac{1}{5}$ = _____ **f)** $\frac{3}{10}$ = _____ **g)** $\frac{2}{5}$ = _____ **h)** $\frac{9}{10}$ = _____



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**6.** Write each percentage as a fraction in its simplest form.

**a)** 50% = _____ **b)** 25% = _____ **c)** 10% = _____ **d)** 75% = _____

**e)** 20% = _____ **f)** 30% = _____ **g)** 1% = _____ **h)** 60% = _____

## D All three together ●●○

**7.** Complete the table. Write each fraction in its simplest form.

| Fraction | Decimal | Percentage |
|---|---|---|
| $\frac{1}{2}$ | | |
| | 0.25 | |
| | | 10% |
| $\frac{3}{4}$ | | |
| | 0.2 | |
| | | 70% |

## E Which is larger? ●●●

⚠ **Watch out:** Do not compare the digits you can see. Change both to percentages first.

**8.** Which is larger?

**a)** $\frac{1}{2}$ or $0.4$ **b)** $30\%$ or $\frac{1}{4}$ **c)** $0.7$ or $65\%$

**d)** $\frac{3}{4}$ or $80\%$ **e)** $0.09$ or $\frac{1}{10}$ **f)** $\frac{1}{5}$ or $25\%$

## F Problems ●●●

**9.** Aroha gets $\frac{7}{10}$ of a test right. Ben gets 65% of the same test right. Who did better? Show your working.

**10.** $\frac{1}{4}$ of the students in a class wear glasses. What percentage of the class do not wear glasses?

## ★ Extension

**11.** Write $\frac{13}{20}$ as a percentage. Start by changing it into hundredths.

**12.** A square is split into 100 small squares and 35 of them are shaded. What fraction of the square is shaded? Give your answer in its simplest form.



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# Fractions, Decimals and Percentages

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**KEY WORDS** fraction *(hautanga)* · decimal *(tau ā-ira)* · percentage *(ōrau)* · per cent *(ōrau)* · tenths *(hautekau)*

## A Tenths and hundredths ●○○

⚠ **Watch out:** The first decimal place is tenths and the second is hundredths: $\frac{7}{10} = 0.7$ but $\frac{7}{100} = 0.07$.

**1.** Write each fraction as a decimal.

**a)** $\frac{1}{10}$ = _____ **b)** $\frac{9}{10}$ = _____ **c)** $\frac{7}{100}$ = _____ **d)** $\frac{37}{100}$ = _____

**e)** $\frac{3}{10}$ = _____ **f)** $\frac{53}{100}$ = _____ **g)** $\frac{3}{100}$ = _____ **h)** $\frac{91}{100}$ = _____

**2.** Write each decimal as a fraction in its simplest form.

**a)** 0.3 = _____ **b)** 0.9 = _____ **c)** 0.07 = _____ **d)** 0.59 = _____

**e)** 0.5 = _____ **f)** 0.4 = _____ **g)** 0.25 = _____ **h)** 0.75 = _____

## B Percentages ●●○

💡 **Remember:** Per cent means "out of 100". 37% is 37 out of 100, which is $\frac{37}{100}$ or 0.37.

**3.** Write each percentage as a decimal.

**a)** 75% = _____ **b)** 40% = _____ **c)** 90% = _____ **d)** 6% = _____

**e)** 88% = _____ **f)** 20% = _____ **g)** 2% = _____ **h)** 35% = _____

**4.** Write each decimal as a percentage.

**a)** 0.35 = _____ **b)** 0.9 = _____ **c)** 0.04 = _____ **d)** 0.6 = _____

**e)** 0.07 = _____ **f)** 0.95 = _____ **g)** 0.5 = _____ **h)** 0.75 = _____

## C Fractions and percentages ●●○

💡 **Remember:** Learn these: $\frac{1}{2} = 50\%$, $\frac{1}{4} = 25\%$, $\frac{3}{4} = 75\%$, $\frac{1}{10} = 10\%$ and $\frac{1}{5} = 20\%$.

**5.** Write each fraction as a percentage.

**a)** $\frac{1}{4}$ = _____ **b)** $\frac{1}{2}$ = _____ **c)** $\frac{1}{5}$ = _____ **d)** $\frac{1}{10}$ = _____

**e)** $\frac{3}{4}$ = _____ **f)** $\frac{7}{10}$ = _____ **g)** $\frac{3}{5}$ = _____ **h)** $\frac{4}{5}$ = _____



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**6.** Write each percentage as a fraction in its simplest form.

**a)** 25% = _____ **b)** 50% = _____ **c)** 20% = _____ **d)** 10% = _____

**e)** 75% = _____ **f)** 70% = _____ **g)** 1% = _____ **h)** 40% = _____

## D All three together ●●○

**7.** Complete the table. Write each fraction in its simplest form.

| Fraction | Decimal | Percentage |
|---|---|---|
| $\frac{1}{4}$ | | |
| | 0.5 | |
| | | 20% |
| $\frac{1}{10}$ | | |
| | 0.75 | |
| | | 30% |

## E Which is larger? ●●●

⚠ **Watch out:** Do not compare the digits you can see. Change both to percentages first.

**8.** Which is larger?

**a)** $\frac{1}{2}$ or $0.6$ **b)** $20\%$ or $\frac{1}{4}$ **c)** $0.8$ or $75\%$

**d)** $\frac{3}{4}$ or $70\%$ **e)** $0.08$ or $\frac{1}{10}$ **f)** $\frac{1}{5}$ or $15\%$

## F Problems ●●●

**9.** Aroha gets $\frac{4}{5}$ of a test right. Ben gets 85% of the same test right. Who did better? Show your working.

**10.** $\frac{1}{5}$ of the students in a class wear glasses. What percentage of the class do not wear glasses?

## ★ Extension

**11.** Write $\frac{9}{20}$ as a percentage. Start by changing it into hundredths.

**12.** A square is split into 100 small squares and 45 of them are shaded. What fraction of the square is shaded? Give your answer in its simplest form.



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